

|          |                                                                |
|----------|----------------------------------------------------------------|
| EX No:1B | <b>EDA – Data Science Frameworks (CRISPDM,<br/>SEMMA, KDD)</b> |
|          |                                                                |

### Aim:

To study Exploratory Data Analysis and apply the CRISP-DM, SEMMA, and KDD data science frameworks to a Life Expectancy dataset by performing data understanding, preprocessing, transformation, modeling, evaluation, and interpretation.

### Algorithm:

#### Algorithm 1: CRISP-DM

1. Import pandas, numpy, and the required machine learning libraries.
2. Upload and load the Life Expectancy Data.csv dataset and inspect the first few records and dataset shape.
3. Remove rows with missing values in Life expectancy, drop the Country column, encode categorical variables, and handle missing feature values using mean imputation.
4. Standardize the feature values using StandardScaler and split the dataset into training and testing sets.
5. Train a Random Forest Regressor model to predict Life expectancy and evaluate it using MAE, MSE, RMSE, and R<sup>2</sup> Score.
6. Apply the CRISP-DM, SEMMA, and KDD frameworks to perform data preparation, modeling, assessment, and interpretation of the Life Expectancy dataset.

### Program:

```
import pandas as pd
import numpy as np
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Upload dataset
uploaded = files.upload()
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
df.columns = df.columns.str.strip()

print("Dataset Loaded Successfully!")
print(df.head())
print(df.shape)

# =====
# CRISP-DM
# =====
crisp_df = df.copy()
```

```

# Business Understanding
# Objective: Predict Life Expectancy

# Data Understanding
print("\n===== CRISP-DM DATA INFORMATION =====")
print("Dataset Shape:", crisp_df.shape)
print("\nMissing Values:")
print(crisp_df.isnull().sum())

# Data Preparation
crisp_df = crisp_df.dropna(subset=["Life expectancy"])

if "Country" in crisp_df.columns:
    crisp_df.drop("Country", axis=1, inplace=True)

le = LabelEncoder()
for col in crisp_df.select_dtypes(include=["object"]).columns:
    crisp_df[col] = le.fit_transform(crisp_df[col].astype(str))

X = crisp_df.drop("Life expectancy", axis=1)
y = crisp_df["Life expectancy"]

imputer = SimpleImputer(strategy="mean")
X = imputer.fit_transform(X)

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

# Modeling
model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

# Evaluation
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== CRISP-DM RESULTS =====")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE :", rmse)
print("R2 Score :", r2)

# =====
# SEMMA
# =====
semma_df = df.copy()

# Sample
semma_df = semma_df.dropna(subset=["Life expectancy"])

# Explore
print("\n===== SEMMA DATA INFORMATION =====")
semma_df.info()

# Modify
if "Country" in semma_df.columns:
    semma_df.drop("Country", axis=1, inplace=True)

le = LabelEncoder()
for col in semma_df.select_dtypes(include=["object"]).columns:

```

```

        semma_df[col] = le.fit_transform(semma_df[col].astype(str))

X = semma_df.drop("Life expectancy", axis=1)
y = semma_df["Life expectancy"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

imputer = SimpleImputer(strategy="mean")
X_train = imputer.fit_transform(X_train)
X_test = imputer.transform(X_test)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Model
model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

# Assess
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== SEMMA RESULTS =====")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE :", rmse)
print("R2 Score :", r2)

# =====
# KDD
# =====
kdd_df = df.copy()

# Step 1: Selection
kdd_df = kdd_df.dropna(subset=["Life expectancy"])

if "Country" in kdd_df.columns:
    kdd_df.drop("Country", axis=1, inplace=True)

# Step 2: Preprocessing
le = LabelEncoder()
for col in kdd_df.select_dtypes(include=["object"]).columns:
    kdd_df[col] = le.fit_transform(kdd_df[col].astype(str))

X = kdd_df.drop("Life expectancy", axis=1)
y = kdd_df["Life expectancy"]

imputer = SimpleImputer(strategy="mean")
X = imputer.fit_transform(X)

# Step 3: Transformation
scaler = StandardScaler()
X = scaler.fit_transform(X)

# Step 4: Data Mining
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

```

```
# Step 5: Interpretation
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("\n===== KDD RESULTS =====")
print("MAE :", mae)
print("MSE :", mse)
print("RMSE :", rmse)
print("R2 Score :", r2)
```

## Output:

Dataset Loaded Successfully!

|   | Country     | Year | Status     | Life expectancy | Adult Mortality \ |
|---|-------------|------|------------|-----------------|-------------------|
| 0 | Afghanistan | 2015 | Developing | 65.0            | 263.0             |
| 1 | Afghanistan | 2014 | Developing | 59.9            | 271.0             |
| 2 | Afghanistan | 2013 | Developing | 59.9            | 268.0             |
| 3 | Afghanistan | 2012 | Developing | 59.5            | 272.0             |
| 4 | Afghanistan | 2011 | Developing | 59.2            | 275.0             |

|   | infant deaths | Alcohol percentage | expenditure | Hepatitis B | Measles ... \ |
|---|---------------|--------------------|-------------|-------------|---------------|
| 0 | 62            | 0.01               | 71.279624   | 65.0        | 1154 ...      |
| 1 | 64            | 0.01               | 73.523582   | 62.0        | 492 ...       |
| 2 | 66            | 0.01               | 73.219243   | 64.0        | 430 ...       |
| 3 | 69            | 0.01               | 78.184215   | 67.0        | 2787 ...      |
| 4 | 71            | 0.01               | 7.097109    | 68.0        | 3013 ...      |

|   | Polio | Total expenditure | Diphtheria | HIV/AIDS | GDP        | Population \ |
|---|-------|-------------------|------------|----------|------------|--------------|
| 0 | 6.0   | 8.16              | 65.0       | 0.1      | 584.259210 | 33736494.0   |
| 1 | 58.0  | 8.18              | 62.0       | 0.1      | 612.696514 | 327582.0     |
| 2 | 62.0  | 8.13              | 64.0       | 0.1      | 631.744976 | 31731688.0   |
| 3 | 67.0  | 8.52              | 67.0       | 0.1      | 669.959000 | 3696958.0    |
| 4 | 68.0  | 7.87              | 68.0       | 0.1      | 63.537231  | 2978599.0    |

|   | thinness 1-19 years | thinness 5-9 years | Income composition of resources \ |
|---|---------------------|--------------------|-----------------------------------|
| 0 | 17.2                | 17.3               | 0.479                             |
| 1 | 17.5                | 17.5               | 0.476                             |
| 2 | 17.7                | 17.7               | 0.470                             |
| 3 | 17.9                | 18.0               | 0.463                             |
| 4 | 18.2                | 18.2               | 0.454                             |

|   | Schooling |
|---|-----------|
| 0 | 10.1      |
| 1 | 10.0      |
| 2 | 9.9       |
| 3 | 9.8       |
| 4 | 9.5       |

[5 rows x 22 columns]  
(2938, 22)

===== CRISP-DM DATA INFORMATION =====

Dataset Shape: (2938, 22)

Missing Values:

|                                 |     |
|---------------------------------|-----|
| Country                         | 0   |
| Year                            | 0   |
| Status                          | 0   |
| Life expectancy                 | 10  |
| Adult Mortality                 | 10  |
| infant deaths                   | 0   |
| Alcohol                         | 194 |
| percentage expenditure          | 0   |
| Hepatitis B                     | 553 |
| Measles                         | 0   |
| BMI                             | 34  |
| under-five deaths               | 0   |
| Polio                           | 19  |
| Total expenditure               | 226 |
| Diphtheria                      | 19  |
| HIV/AIDS                        | 0   |
| GDP                             | 448 |
| Population                      | 652 |
| thinness 1-19 years             | 34  |
| thinness 5-9 years              | 34  |
| Income composition of resources | 167 |
| Schooling                       | 163 |

dtype: int64

===== CRISP-DM RESULTS =====

MAE : 1.0588054607508561  
MSE : 2.7685972969283315  
RMSE : 1.6639102430504873  
R2 Score : 0.9679928736778992

===== SEMMA DATA INFORMATION =====

<class 'pandas.core.frame.DataFrame'>  
Index: 2928 entries, 0 to 2937

```

===== SEMMA DATA INFORMATION =====
<class 'pandas.core.frame.DataFrame'>
Index: 2928 entries, 0 to 2937
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Country                               2928 non-null   object
1   Year                                  2928 non-null   int64
2   Status                                2928 non-null   object
3   Life expectancy                       2928 non-null   float64
4   Adult Mortality                       2928 non-null   float64
5   infant deaths                         2928 non-null   int64
6   Alcohol                               2735 non-null   float64
7   percentage expenditure                 2928 non-null   float64
8   Hepatitis B                           2375 non-null   float64
9   Measles                               2928 non-null   int64
10  BMI                                    2896 non-null   float64
11  under-five deaths                     2928 non-null   int64
12  Polio                                 2909 non-null   float64
13  Total expenditure                     2702 non-null   float64
14  Diphtheria                            2909 non-null   float64
15  HIV/AIDS                              2928 non-null   float64
16  GDP                                    2485 non-null   float64
17  Population                             2284 non-null   float64
18  thinness 1-19 years                   2896 non-null   float64
19  thinness 5-9 years                    2896 non-null   float64
20  Income composition of resources       2768 non-null   float64
21  Schooling                             2768 non-null   float64
dtypes: float64(16), int64(4), object(2)
memory usage: 526.1+ KB
None

```

```

===== SEMMA RESULTS =====
MAE : 1.0500972696245772
MSE : 2.768727247440278
RMSE : 1.6639492923284285
R2 Score : 0.9679913713494616

```

```

===== KDD RESULTS =====
MAE : 1.0588054607508561
MSE : 2.7685972969283315
RMSE : 1.6639102430504873
R2 Score : 0.9679928736778992

```

**Result:**